

MOUHSSINE EL BOUMSHOULI

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Summary

AI engineering student at EIDIA, Euromed University of Fes, building full-stack, AI-assisted, and computer-vision systems with Python, FastAPI, React, PostgreSQL, and Docker — with hands-on experience taking a product from concept to paying customers as an independent founder.

Experience

Software & AI Engineering Intern — Bounaim Auto, Beni Mellal, Morocco Jul – Aug 2026

- Designed and delivered SmartImport, a full-stack procurement platform for supplier quotations, cost analysis, and AI-assisted document extraction.
- Built completeness-aware supplier comparison, Decimal-based landed-cost logic, staged PDF/XLSX ingestion, and Dockerized delivery.
- Validated the system with 139 backend tests, reaching 98.99% document-field F1 and 92.31% top-1 matching accuracy.

Founder & Developer — Convoroute LLC, United States (Remote) Jun 2026 – Present

- Building a B2B SaaS for embedded AI chatbots supporting customer service, FAQ automation, and lead capture.
- Lead product direction across LLM integration, Stripe billing, and deployment for a self-service client platform.
- Own client onboarding and delivery end to end, from initial configuration through ongoing support.

Projects

DARE-Bench Agent Reliability Study — Python, OpenAI API, pandas, scikit-learn, Docker 2026

- Designed and ran a reproducible 240-run study on a frozen 24-task DARE-Bench subset, comparing 3-turn and 5-turn agent budgets across 5 repeats per task.
- Found exact-match success rising from 34.2% to 50.0%, while flaky-task rate rose from 29.2% to 41.7% and pairwise disagreement from 15.0% to 21.7%.
- Built official rescoring, failure-taxonomy analysis, and task-bootstrap confidence intervals; classified 94 of 139 failed runs as code errors.

SmartImport — Procurement Decision-Support Platform — React, TypeScript, FastAPI, PostgreSQL, SQLAlchemy, Docker 2026

- Engineered end-to-end purchasing workflows spanning requirements, quotations, FX assumptions, and landed-cost analysis.
- Built an auditable architecture with immutable snapshots and staged document ingestion.
- Reached 98.99% document-field F1, 100% product-line F1, and 92.31% top-1 supplier-matching accuracy across 139 backend tests.

medskel — Medical Image Skeletonisation — Python, OpenCV, Shapely, NumPy 2026

- Implemented a polygon/bisector skeletonisation method for 2D medical segmentations, validated with 28 hand-derived tests and 9 reproducible experiments.
- Reduced inter-observer vessel-length variation from 9.3% to 3.2% versus pixel thinning on 28 retinal images ($p < 0.0001$).
- Built the geometric-processing and evaluation pipeline entirely in Python.

UEMF Présence — Smart Attendance System — Next.js, React, PostgreSQL, Prisma, Vercel 2026

- Built a role-based attendance platform with GPS check-in and HMAC-SHA256 QR tokens rotating every 10 seconds.
- Added anti-fraud checks, professor review workflows, and an administration anomaly dashboard.
- Implemented separate student, professor, and administrator workflows, deployed on Vercel.

Intelligent Autocomplete & Query Suggestion Engine — C, Ternary Search Tree 2026

- Built a pure-C autocomplete engine using a ternary search tree, with fuzzy matching and top-k ranking.
- Implemented dynamic dictionary updates and manual memory management, with no external dependencies.
- Added benchmarking and memory-profiling modes to measure lookup speed and memory use.

Education

Engineering Degree in Artificial Intelligence — EIDIA, Euromed University of Fes 2025 – Expected 2028

Preparatory Cycle in Digital Engineering — EIDIA, Euromed University of Fes 2023 – 2025

- Selected coursework: Advanced Algorithms & Data Structures, OOP, Databases, Machine Learning, Probability & Statistics, Cybersecurity.

Skills

Programming: Python, TypeScript, C, C++, Java, SQL, R, MATLAB, HTML/CSS. **AI & Computer Vision:** LLM integration, prompt design, OpenAI APIs, OpenCV, NumPy. **Backend & Web:** FastAPI, SQLAlchemy, Pydantic, PostgreSQL, Alembic, Prisma, React, Next.js, REST APIs. **Tools & Quality:** Docker, Nginx, Git/GitHub, Linux, pytest, Vitest, Ruff, mypy, ESLint, Vercel. **Languages:** Arabic (native), English (C1 — TOEFL iBT 112/120), French (working proficiency), Spanish (beginner).